

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)
(Estd. under the Panjab University Act VII of 1947-enacted by the Govt. of India)

OUTLINES OF TESTS SYLLABI AND COURSES OF READING

FOR

B.Voc.Multimedia (Graphics & Animation)
I, II and IIIrd Year

Session 2018-19 New Scheme (1st - 6th Semester)}

SCHEME OF B.Voc. MULTIMEDIA

(GRAPHICS & ANIMATION)

(SEMESTER SYSTEM)

Semester I								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN -101	Communication Skills	Generic	Theory	20	80	--	--	6
*GEN -102	Fundamentals of Information Technology	Generic	Theory	20	80	--	--	6
MGA - 103	Understanding of Multimedia & Animation	Skill	Theory	20	80	--	--	6
MGA - 104	CorelDraw and Photoshop	Skill	Theory	20	80	--	--	6
MGA - 105	Practical based on Paper MGA -101 & MGA- 102	Skill	Practical	--	--	20	80	6
SEMESTER II								
*GEN 201	Soft Skills and Personality Development	Generic	Theory	20	80	--	--	6
*GEN-202	Programming in C	Generic	Theory	20	80	--	--	6
MGA-203	Cinematography	Skill	Theory	20	80	--	--	6
MGA-204	Video and Audio Editing	Skill	Theory & Practical	10	40	10	40	6
MGA-205	Art Foundation : Drawing	Skill	Practical	--	--	20	80	6
**SIT-201	Summer Training [OJT] After 2nd Semester Examinations	Skill	Practical	--	--	20	80	6

***Refer to Generic Components Common to all B.Voc. Courses**

**** Summer Industrial Training of 4-6 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within in one week of start of 3rd Semester. Viva-Voce examination to be held within 3-weeks of the start of 3rd semester.**

Job Role: Web Designer, 2D Animator, 3d Modeler & Architectural Designer.

Semester III								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN 301	Value Education And Human Rights	Generic	Theory	20	80	--	--	6
*GEN-302	Internet Technology & Web Designing	Generic	Theory	20	80	--	--	6
MGA - 303	Flash	Skill	Theory	20	80	--	--	6
MGA - 304	Dreamweaver	Skill	Theory	20	80	--	--	6
MGA - 305	Practical Based on MGA-303 & MGA-304	Skill	Practical	--	--	20	80	6
SEMESTER IV								
*GEN 401	Environmental Studies	Generic	Theory	20	80	--	--	6
*GEN-402	Animation Theory	Generic	Theory	20	80	--	--	6
MGA-403	3ds Max: Modeling	Skill	Theory	20	80	--	--	6
MGA-404	3ds Max: Texturing, Lighting & Rendering	Skill	Theory	20	80	--	--	6
MGA-405	Practical Based on MGA-403 & MGA-404	Skill	Theory & Practical	--	--	20	80	6
**SIT	Summer Training [OJT] After 2nd Semester Examinations	Skill	Practical	--	--	20	80	6

***Refer to Generic Components Common to all B.Voc. Courses**

**** Summer Industrial Training of 4-6 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within in one week of start of 5th Semester. Viva-Voce examination to be held within 3-weeks of the start of 5th semester.**

Job Role: Java Developer, 3D Animator, VFX Editor.

Semester V								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN - 501	Critical Thinking and Elementary Statistics	Generic	Theory	20	80	--	--	6
*GEN-502	Cyber Laws	Generic	Theory	20	80	--	--	6
MGA-503	3D Max Animation	Skill	Practical	--	--	20	80	6
MGA-504	Visual Effect	Skill	Practical	--	--	20	80	6
MGA-505	Pre-Production	Skill	Theory	20	80	--	--	6

Note: Winter Industrial/ In-house Training of 2-3 weeks in a relevant area after 5th Semester Examinations in winter break.
Training report by the student to be submitted within in one week of start of 6th Semester. Viva-Voce examination to be held within 3-weeks of the start of 6th semester.

SEMESTER VI

Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN 601	Entrepreneurship Development Programme	Generic	Theory	20	80	--	--	6
*GEN-602	Design & Communication Process	Generic	Theory	20	80	--	--	6
MGA-603	Final Project	Skill	Practical	--	--	60	240	18
**SIT-601	Summer Industrial/ In-house Training and Comprehensive Viva	Skill	Practical	--	--	20	80	6

The Project will be 6 months duration.

<u>Specialization (Choose any one Project)</u>

1. **Final Project:** Graphics Designing / Web Designing / 2D Animation / 3D Animation / Texturing & Shading, Lighting & Rendering / Visual Effects / Digital Compositing / Motion Graphics / Video Editing.

Note:

1. In the six semester students will choose any one project to make their own portfolio/Show reel for the specialization.

Job Role: Graphics Designer, Web Designer, Video / Audio Editing, VFX Compositor, 2D Animator, 3D Modular.

B.Voc. Multimedia (Graphics & Animation) Semester-1st

Paper Title: Communication Skills

Paper Code: GEN 101 **Credits: 6**

Job Role: Business Communication.

Objective: The objective of this paper is to develop communication skills, discover what business communication is all about and learn how to adapt the communication experiences in life and to the business world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Introduction to Communication: Need for Effective Communication, The Process of Communication, Levels of communication, Flow of communication, Use of language in communication, Communication networks, Significance of technical communication, Barriers to Communication, Types of barriers, Miscommunication, Noise, overcoming measures.

UNIT -II

Verbal Communication: Planning, Preparation, Delivery, Feedback and Assessment of Activities like –Public Speaking, Group discussion, Presentation Skill, Audio-Visual Aids, Personal interview**. Non-Verbal Communication:** Body Language, Personal Appearance, Posture, Gestures, Facial Expressions, Positive attitude formation, Process of attitude formation, How to build a successful attitude.

UNIT –III

SWOT analysis: Self-management techniques, Self-image and Self-esteem, Building selfconfidence, Power of irresistible enthusiasm, Etiquettes, Etiquettes in Social as well as Office Atmosphere, Telephone Etiquettes, E-mail Etiquettes, etiquettes and manners. Importance of listening and responding.

UNIT -IV

Report Writing: Business reports: Types, Characteristics, Importance, Elements of structure, Process of writing, Order of writing, the final draft, check lists for reports. Business Correspondence: Inviting quotations, Sending quotations, placing orders, Inviting tenders, Sales letters, claim & adjustment letters and social correspondence.

Books Recommended:

1. 'Management: A Global and Entrepreneurial Perspective', by Heinz Weihrich,
2. Mark.V.Cannie, Harold Koontz., Tata McGraw Hill Publication
3. 'Developing Communication Skills' by Krishmohan and Meera Banerjee, Macmilan India Ltd.
4. 'Business Communication' by Krizan, Merrier, Jones, Thomson Learning - 6thEdition
5. 'Communication Skills' by Sanjay Kumar & PushpLata, Oxford University Press
6. 'An Approach to Communication Skills' by Indrajit Bhattacharya, Delhi : Dhanpat Rai
7. 'Introduction to Psychology' by Atkinson and Hilgard's, Edward E. Smith, Susan
8. Nolen-Hoeksema, Barbara Fredrickson, Geoffrey Loftus

B.Voc. Multimedia (Graphics & Animation) Semester-1st

Paper Title: FUNDAMENTALS OF INFORMATION TECHNOLOGY

Paper Code: GEN 102 **Credits: 6**

Job Role: Computer Oprator

Objective: The objective of the paper is to familiarize the students with developments in Information Technology, Internet and use of computer systems at operating system level and application level.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT - I

Computers: Introduction to computers, characteristics of computer, organization of computers, hardware, software, data, information, Types of computer: Classification on the basis of purpose: digital computers, analog computers, hybrid computers; Classification on the basis of size: microcomputers, mini computers, mainframe computers and supercomputers, desktop computers, laptops, workstations, PDA. Generations of computer, Uses and Application of computers.

UNIT- II

Computer Hardware: Input Devices: Keyboard, mouse, light pen, joystick, trackball. Voice input device: microphone; Output Devices Printers: types of printer, Plotters, Speakers; Scanners: types of scanners, Visual display devices. **Computer Storage:** Representation of information, BIT, BYTE, Memory, Types of memory: primary memory RAM: static RAM, Dynamic RAM; ROM, EROM, EPROM, EEPROM; Secondary Memory: Magnetic disk, Hard disk, Floppy disk, Optical disk, Compact disk (CD-ROM) and Solid state storage devices.

UNIT - III

Software: Introduction, Types of Software: Application software, System software. Operating system, functions of operating system, types of operating system. Data processing, Data processing systems: batch processing, online processing, time sharing, real-time applications, Single-user, multi-user, and client-server systems; distributed and parallel processing systems; Translators: compilers, interpreters and assemblers.

UNIT -IV

Computer Networks: Introduction, types of networks on the basis of area coverage: LAN, WAN, MAN.
Internet and WWW: Evolution of Internet, Various Internet services (WWW, e-mail, telnet, ftp, IRC, news) and their uses, Access Methods, Browsers, Future of Internet, Applications of Internet, Evolution of www.

Books Recommended:

1. 'Computers Today' by S.K. Basandra, Galgotia Publications.
2. 'Computer Fundamentals' by P.K.Sinha, N.D.: BPB. Publications.
3. 'Fundamentals of Computers' by V. Rajaraman, N.D.; PHI Publications.
4. 'Using the Internet' by Barbara Kasser, PHI, 4th ed., New Delhi.
5. 'Using the World Wide Web' by David A. Wall, PHI, New Delhi.

B.Voc. Multimedia (Graphics & Animation) Semester-1st

Paper Title: Understanding of Multimedia & Animation

Paper Code: MGA-103 **Credits: 6**

Job Role: Multimedia Editor

Objective: The objective of the paper is to familiarize the students with animation history and multimedia tools.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Introduction to Multimedia: Multimedia and Hypermedia, Overview of Multimedia Software Tools. Authoring Tools-Multimedia Authoring, Editing and Authoring Tools.

Graphics and Image Data Representations: Graphics/Image Data Types, File Formats-GIF, JPEG, PNG, TIFF, EXIF, PDF, WMF, BMP.

Color in Image and Video: Color Science, Color Models in Images-RGB, CMY, CMYK, Color Models in Video.

UNIT II

Fundamental Concepts in Video: Types of Video Signals, Analog Video-NTSC, PAL, SECAM, Digital Video.

Basics of Digital Audio: Digitization of Sound, MIDI, Quantization and Transmission of Audio. Multimedia Data Compression: Lossless Compression-Run Length Encoding, Variable Length Coding, Dictionary Based Coding, Arithmetic Coding. Lossy Compression Algorithms-Distortion Measures, Rate Distortion Theory, Quantization, Transform Coding, Wavelet Coding.

UNIT III

Multimedia Communication & Retrieval: Computer and Multimedia Networks-Basics of Computer and Multimedia Networks, Multiplexing Technologies, LAN & WAN, Access Networks, Common Peripherals Interfaces.

Wireless Networks: Analog Wireless Networks, Digital Wireless Networks, TDMA GSM, CDMA, 3G, Wireless LAN. Radio Propagation Models, Multimedia Over Wireless Networks.

UNIT IV

History of Animation, Types of Animation, Skills for Animation Artist, Basic Principles of Animation, Animator's Drawing Tools, Rapid Sketching & Drawing, Developing Animation Character, Essentials & Qualities of Good Animation Characters, Three Dimensional Drawings of Characters.

References:

1. Ze-Nian li, Mark S. Drew,2004:*Fundamentals of Multimedia*, PHI.
2. Chris Patmore , 2009, *The Complete Animation Course* , Barons Educational Series
3. Vaughan, Tay, 2008 : *Multimedia Making it Work*, Osborne Publishers.
4. Judith Jeffcoate, 2007 : *Multimedia in Practice, Technology and Applications*, PHI.

B.Voc. Multimedia (Graphics & Animation) Semester-1st

Paper Title: CorelDraw and Photoshop

Paper Code: MGA-104 **Credits: 6**

Job Role: Graphics Designer.

Objective: The objective of the paper is to familiarize the students with developments in graphics designing.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Introduction to CorelDraw: CorelDraw Screen, Drawing Lines, Selecting Objects, Creating Artistic Text, Formatting Text, Working with Shapes, Working with Rectangles, Creating Ellipses and Circles, Drawing Polygons, Drawing Spirals, Editing and Transforming Shapes, Saving Files, Printing Files.

Controlling CorelDraw Environment: Viewing and Moving Toolbars, Working with Docker Window, Setting up Page Layout, Defining Outlines.

UNIT-II

Working with Shapes and Curves: Drawing and Editing Freehand Curves, Editing Shapes, Editing Curves, Working with Bitmap Images in CorelDraw, Applying Effects to Bitmaps.

PHOTO-PAINT and 3D Basics: Creating a New Bitmap Image, Setting Colors and Fills, Painting Text , Painting Shapes, General 3D Concepts.

UNIT-III

Adobe Photoshop Basics: Introducing Adobe Photoshop, Exploring the Photoshop Workspace, Touring the Workspace, Exploring the Photoshop Menu Bar.

Image Basics: Exploring File Types, Understanding Image Files, Understanding Video Files, Creating and Opening Images, Resizing Files and Adjusting Resolution, Understanding Colors Basics, Adjusting with Histograms, Working in Different Color Modes.

UNIT-IV

Creating Selections: Using the Selection Tools, Working with Lasso Tools and Quick Selection Tools, Refining and Adjusting Selection.

Applying Sharpness and Blur Adjustment: Using Sharpening Filter, Using Blur Filter, Introducing the Blur Gallery, Using the Healing Brush Tools

References :

1. Dave Karlins,1999:*Sam'sTeach Yourself CorelDraw9 in 24 hours,* Sams Publishing.
2. Brad Dayley & DaNae Dayley,2012:*Adobe Photoshop CS6 BIBLE,* Wiley.
3. Altman, 1999 : *Mastering Corel-Draw,* BPB.
4. Gary David Bouton, 2012: *CorelDRAW X6 The Official Guide*, McGraw Hill Professional.
5. Romaniello, 2002 : *Photoshop 7 – Savvy*, Sybex.
6. Adobe Creative Team, 2012: *Adobe Photoshop CS6 Classroom in a Book*, Adobe Press.

B.Voc. Multimedia (Graphics & Animation) Semester-1st

Paper Title: Practical based on Paper MGA-103 & MGA-104

Paper Code: MGA-105 **Credits: 6**

Job Role: Multimedia Editor & Graphics Designer

There will be a practical examination in this paper along with semester examination. External examiner will evaluate the skills of the students in the field of CorelDraw and Photoshop. Examiner will give on-the-spot assignments to the students. Besides this a viva voce (oral examination) of each student one by one will also be taken by the examiner for testing the knowledge of the students.

B.Voc. Multimedia (Graphics & Animation) Semester- 2nd

Paper Title: SOFT SKILLS AND PERSONALITY DEVELOPMENT

Paper Code: GEN-201 **Credits: 6**

Job Role: Creative Salesmanship

Objective: To expose the students to the concept of 'Human Development' (Personal and interpersonal) with emphasis on the latent resources that every human being possesses. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Attitude, Process of attitude formation, How to build a success attitude, Spot analysis, Selfmanagement techniques, Self-image and self-esteem, Building self -confidence, Power of irresistible enthusiasm, etiquettes and manners in a group, public speaking, oral and written communication, Body language, Importance of listening and responding, tips for technical writing.

UNIT - II

Development of Communication & Co-operation. Functions of Communication, Communication Basics, Communication Networks, Tips for Effective Internal Communication. Non-verbal Communication. Ethical Communication: austerity in speech, value, ethics and communication. Communication aids. Suitable behavior towards customers, Influence in skill, Creativity in presentation & projection and Multi-cultural skills

UNIT -III

Introduction to Personality-Basic of Personality, Human growth and Behavior, Theories in Personality, Motivation; Techniques in Personality development – Self-confidence, Mnemonics, Goal setting, Time Management and effective planning, Techniques in Personality Development-Stress Management, Meditation and concentration techniques, Selfhypnotism, Self-acceptance and Self-growth.

UNIT -IV

Co-ordination while working in a team, Leadership styles, Leader & Team player, Management of conflict, Profiles of great and successful personalities, Role of career planning in personality development, How to face personal interviews and group discussions.

Books Recommended:

1. 'Personality Development' by Rajiv K. Mishra, Rupa & Co
2. 'An Approach to Communication Skills' by Indrajit Bhattacharya, Delhi : Dhanpat Rai
3. 'Business Communication Skills' by Varinder Kumar, Bodh Raj, Manocha, Kalyani Publishers, New Delhi.
4. 'Introduction to Psychology' by Atkinson and Hilgard's ,Edward E. Smith, Susan
5. Nolen-Hoeksema, Barbara Fredrickson, Geoffrey Loftus
6. 'Communication Today & Tomorrow' by Ravi Aggarwal, Sublime Publications, Jaipur

B.Voc. Multimedia (Graphics & Animation) Semester- 2nd

Paper Title: Programming in C

Paper Code: GEN-202 **Credits: 6**

Job Role: C Programmer.

Objective: To expose the students to the concept of C language programmer. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Fundamentals of 'C': I/O statements, assignment statements, Constants, variables and data types, Operators and Expressions, Standards and Formatted statements, Key word, Data Types and Identifiers.

UNIT II

Control Structures: Introduction, Decision making with IF – statement, IF – Else and Nested IF, While and do-while, for loop, Jump statements, break, continue, goto, Switch Statement. Functions: Introduction to functions, Global and Local Variables, Function Declaration, Standard functions..

UNIT III

Preprocessor Directives: Introduction and Use, Macros, Conditional Preprocessors, Header Files, Arrays: Introduction to Arrays, Array Declaration, Single and Multidimensional Array, Arrays of characters
Pointers: Introduction to Pointers, Address operator and pointers, Declaring and Initializing pointers.

UNIT IV

Structures and Unions: Declaration of structures, Accessing structure members, Structure Initialization, Arrays of structure, nested structures, structure with pointers, Unions.
Strings: Introduction, Declaring and Initializing string variables, Reading and writing strings, String handling functions.

References:

1. Byron S. Gottfried, 1996 : Programming in C, McGraw Hills Publishers, New York.
2. Salaria, R. S. : Test Your Skills in C, Khanna Book Publishing Co. (P.) Ltd.,New Delhi.
3. Kanetkar Yashwant, 2010 : Let us Exploring C, BPB Publications, New Delhi
4. Balaguruswami C., 2008 : Programming with C Language, Tata McGraw Hill, New Delhi.
   Somashekara, M.T. , 2008 : Programming in C, Prentice Hall of India.

B.Voc. Multimedia (Graphics & Animation) Semester- 2nd

Paper Title: Cinematography

Paper Code: MGA-203 **Credits: 6**

Job Role: Cameraman, Film Maker.

Objective: To expose the students to the concept of camera angles. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Writing with Motion: Conceptual Tools of Cinematography, Frame, Lens, Light and Color, Texture, Movement, Establishment, Point-of-View.

Shooting Methods: Cinematic, The Frame, Cinema as a Language, Shooting Methods.

Visual Language: Design Principles, Forces of Visual Organization, Rules of Composition.

Language of the Lens: Lens and the Frame, Image Control at the Lens.

UNIT II

Visual Storytelling: Visual Metaphore, Lighting as Storytelling, Lighting as Visual Metaphore.

Cinematic Continuity: Shooting for Editing, Prime Directive, Screen Direction, Turnaround, Planning Coverage, Issues in Continuity and Shooting, Types of Cuts.

Lighting Basics: Fundamentals of Lighting, Lighting Terminology, Aspects of Light, Basic Lighting Techniques, Motivated Light, Day Exteriors, Lighting for High Def. Video.

Lighting Sources: Tools of Lighting, Daylight Source, Xenons, PARs, Soft Lights, Color Correction Fluorescents.

UNIT III

HD Cinematography: High Def. and Standard Def. Videos, Analog & Digital Video, Types of Video Sensors, Digital Video, Shooting Formats, Digital Compression, Monitoring on the Set, Video Signal, Setting Up a Color Monitor, Camera with Balance, Timecode & Edgecode, Compression & Codecs.

Exposure: Light as Energy, Exposure, ISO, and Lighting Relationships, Light & Film, Densitometry, Brightness Perception, Contrast, Zone System, Gray Scale, Exposure & Camera.

Camera Movement: Basic Techniques, Moves, Moving Shots, Camera Mounting, Crab Dolly, Cranes, Car Shots, Ariel Shots, Motion Control.

UNIT IV

Image Control: Color Printing, Controlling Color & Contrast, LookUp Tables, Camera Filter Types, Color Temperature & Filtration, Image Control with Camera.

Optics & Focus: Basic of Optics, Focus, Depth-of-Field, Macrophotography.

Set Operations: Director of Photography, Team, Time Code Slates.

Technical Issues: Flicker, Filming Practical Monitors, Shooting Process, Dimmers, High Speed Photography, Underwater Filming, Image Quality, Effects, Sun Location with Compass, Transferring Film to Video**,** Film Formats.

References:

1. Blain Brown, 2013: *Cinematography, Theory And Practice*, Focal Press.
2. Kris Malkiewicz, M. David Mullen, 2009: *Cinematography: Third Edition*, Simon and Schuster.
3. Joseph v. Mascelli, 1998: *The Five C's of Cinematography,* Silman-James Press.
4. Viktor Shklovkii, 2008: *Literature and Cinematography.* Dalkley Archive Press.

B.Voc. Multimedia (Graphics & Animation) Semester- 2nd

Paper Title: Video and Audio Editing

Paper Code: MGA-204 **Credits: 6**

Job Role: Video & Audio Editor

Objective: To expose the students to the concept of video & audio editor. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Introduction to Video Editing, Non-Linear Editing: Meaning and Process, Hardware Requirements for NLE, Connecting Editing Equipment, Creating Time Line, Cut, Fade, Mix and Wipe, Introduction to Editing Aesthetics.

Unit II

Main Tools of Editing - Selection, Range Select, Rolling, Edit, Razor, Hand Tool, Cross Fade, In Point, Zoom Tools , Video Transitions, Titling and Graphics, Rendering and Authoring, Giving Effects, Mixing Video, Editing on The Tape, Hard Disk.

Unit III

Nature of Digital Sound, Meaning of Mono, Stereo and Surround Sound, Various Formats of Digital Sound, Editing Tools: Cut, Cross Fade, Mix, Zero-Crossing Edits, Multiple Track Editing Process of Equalization, Applying Effects to Sound.

Unit IV

Introduction to Audio Console, Audio Mixing, Deleting Unwanted Audio, Getting Output, Fades and Cross-Fades, Giving Audio Effects, Editing Sound Using Tape, Hard Disk, Introduction to Sound Aesthetics, Introduction to Sound Forge, Nuendo

References:

1. Jhon Rice and Brian Mckernan, 2003: *Editing Digital Video,* McGraw-Hill Companies
2. Ken Dancyger, 2011:*The Technique of Film and Video Editing: History, Theory, and Practice*, Focal Press .
3. Simon Langford, 2013:*Digital Audio Editing*, Focal Press.
4. Sam McGuire, David Liban,2010:*The Video Editor's Guide to Soundtrack Pro: Workflow, Tools and Techniques*, Focal Press.
5. Dave D. Peck, 2012: *Video Editing with Adobe Premiere*, Cengage Learning.
6. Scott R. Garrigus, 2010: *Sound Forge Power*, Firewall Media.

B.Voc. Multimedia (Graphics & Animation) Semester- 2nd

Paper Title: Art Foundation: Drawing

Paper Code: MGA-205 **Credits: 6**

Job Role: Cartoonist, Caricature Artist, Drawing Artist.

Objective: To expose the students to the concept of cartoonist. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Free Hand Sketching From Real Objects: Building, Vehicles, Chair, Table, Trees etc. Preparing Colour Charts. Kind of Design- 2 D & 3 D Design

UNIT II

Designing Letter Heads and Visting Cards, Designing Book Covers and Folders.

UNIT III

Landscape on the Spot & Landscape Form Memory (Hills, Trees, Hut, Rivers etc.), Layout Designing of Newspapers, Magazines, Print Advertisements.

UNIT IV

Poster Making on Various Themes and Designing of Hoarding Making.

Note: There will be a practical examination in this paper along with semester examination. External examiner will evaluate the skills of the students in the field of drawing. Examiner will give on-the-spot assignments/tasks to the students. Besides this a viva voce (oral examination) of each student one by one will also be taken by the examiner for testing the knowledge of the students.

Books Recommended:

- Drawing and sketching in pencil (Author : L.Guptill)
- Draw How to Master the Art (Jeffery Camp) Foreword by David Hockney.
- The Art of Drawing (Willy Pongany)
- Prachi- Visual Arts.

B.Voc. Multimedia (Graphics & Animation) Semester- 3rd

Paper Title: VALUE EDUCATION AND HUMAN RIGHTS

Paper Code: GEN 301 **Credits: 6**

Job Role: Holistic Marketing

Objective: The objective of this paper is to impart basic human values to students through formal education and contribute to making the student a true human being, who is able to face life and make it meaningful.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Value Education : Its purpose and significance in the present world ; Value system – The role of culture and civilization; Holistic living – Balancing the outer and inner –Body, Mind and Intellectual level- Duties and responsibilities.

UNIT II

Salient values for life- Truth, commitment, honesty and integrity, forgiveness and love, empathy and ability to sacrifice, care, unity , and inclusiveness, Self-esteem and selfconfidence, punctuality – Time, task and resource management – Problem solving and decision making skills- Interpersonal and Intra personal relationship – Team work – Positive and creative thinking.

UNIT III

Human Rights – Universal Declaration of Human Rights – Human Rights violations –National Integration – Peace and non-violence – Dr. A P J Kalam's ten points for enlightened citizenship – Social Values and Welfare of the citizen – The role of media in value building.

UNIT IV

Social Evils – Corruption, Cyber crime, Terrorism – Alcoholism, Drug addiction – Dowry –Domestic violence – untouchability – female infanticide – atrocities against women- How to tackle them.

Books Recommended:

1. 'Education and Human Values' by M.G.Chitakra, A.P.H. Publishing Corporation, New Delhi, 2003
2. 'Values and Ethics for Organizations: Theory and Practice' by Chakravarthy, S.K. ,Oxford University Press, NewDelhi , 1999.
3. 'Ethics, Education, Indian Unityand Culture' by Satchidananda, M.K., Ajantha Publications, Delhi, 1991
4. 'Social Values among Youngadults: A changing Scenario' by Das, M.S. & Gupta, V.K. , M.D. Publications, NewDelhi, 1995.
5. 'Humanist Values: A Source Book' by Bandiste, D.D., B.R.Publishing Corporation, Delhi, 1999.
6. 'Human Values and education' by Ruhela, S.P. , SterlingPublications, New Delhi, 1986.
7. 'Values and Education in Independent Indian' by Kaul, G.N.,Associated Publishers, Mumbai, 1975
8. 'Education in Values', NCERT, New Delhi, 1992.

B.Voc. Multimedia (Graphics & Animation) Semester- 3rd

Paper Title: Internet Technology& Web Designing

Paper Code: GEN 302 **Credits: 6**

Job Role: Web designer.

Objective: To expose the students to the concept of web designer. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Introduction to Internet: Internet Overview, Evolution & History of Internet, Growth of Internet, Internet Services, Anatomy of Internet, Internet Applications, Impact of Internet.

Internet Technology and Protocol: OSI Reference Model, TCP/IP Reference Model, Data Transmission, Switching, Routers and Gateways, Network Protocols, Internet Addressing.

Internet Connectivity: Different Types of Connections, Hardware and Software requirements of Internet, Levels of Internet Connectivity, Internet Service provider.

UNIT-II

Computer Network: Network Definition, Common Terminology- LAN, WAN, Network Components, Network Topologies, Types of Network Architecture, Different Medium of Communication, Network Security.

WWW and Web browser: WWW, Evolution of Web, Basic Elements of WWW, Web Browsers and Search Engines.Electronic Mail: Email Basics, Concepts of Email Server, Structure of Email Message, Working of Email System, Email Security.

UNIT-III

Web Publishing and Browsing: Overview, SGML, Web Hosting, URL, CGL, Components of web Publishing, Web Page Design consideration and Principles.

HTML Programming Basics: HTML Page Structure, HTML Lists, HTML Tables, HTML Links, HTML Forms, HTML Frames.

UNIT-IV

Internet Tools and Multimedia: Current Trends on Internet, Interactivity Tools, Multimedia and Animation.

Remote Login: introduction to Remote login, Introduction to Telnet, File Transfer Protocol.

Internet and Web Security: Overview of Internet Security, Email Threats, Web Security and Privacy Concepts, Firewall, Cryptography, Digital Signatures, Authentication and Access Control, Virus, Internet Security and Management Concepts.

References:

1. ISRD Group, 2011: *Internet Technology and Web Design*, Tata McGraw-Hill.
2. Tom Craig, 2003: *Internet: Technology, People, Process*, Black Rabbit Books.
3. Valerie Bodden, 2008: *Internet,* The Creative Company.
4. Ewaryst Tkacz, Adrian Kapczynski, 2009: *Internet-Technical Development and Applications,* Springers Publications.

B.Voc. Multimedia (Graphics & Animation) Semester- 3rd

Paper Title: Flash

Paper Code: MGA-303 **Credits: 6**

Job Role: 2D Animator.

Objective: To expose the students to the concept of 2D Animation. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Understanding the Adobe Flash: The Topography of Flash CS5, Bitmap Handlers, Vector Based Drawing Programs, Flash in Internet Evolution, Exploring Companion Technologies, Planning Flash Projects.

Mastering in Flash Environment: Interface Fundamentals, Managing Windows and Panels, The Tools Panel, Document and Timeline Window, Drawing in Flash, Using Shapes and Drawing Tools, Using Fill and Stroke Controls, Understanding Symbols Instances and Library.

UNIT –II

Applying Color and Graphics: Introducing Color basics, Working in the Swatches Panel,Using the Color Panel, Modifying Graphics, Transforming Gradients and Bitmap Fills, Manipulating Symbols in 3D Space.

Integrating Media Files with Flash: Adding Sound, Importing Sound in Flash, Synchronizing Audio to Animation, Stopping Sounds, Editing Audio in Flash.

UNIT-III

Defining Vector, Bitmaps and Video: Importing and Copying Bitmaps, Setting Bitmap Properties, Bitmap Compression, Converting Raster to Vector.

Displaying Video: Integrating Video, Importing the Video, Compressing Video with Adobe Media Encoder CS5.

Adding Basic Interactivity to Flash Movie: Understanding Action and Event Handlers, First Five Actions, Building Timelines and Interactions.

UNIT-IV

Distributing Flash Movies: Testing Flash Movies, Publishing your Flash Movies, Selecting Formats, Integrating Flash Content with Web Pages, Writing Markup for Flash Movies, Detecting the Flash Player. Using The Flash Player: Using the Stand-Alone Flash Player, Using the Flash Player Plug In for Web Browsers, Using the Flash Player Setting.

References:

1. Todd Perkins, 2011: *Adobe Flash Professional CS5, Bible*, Wiley India.
2. Adobe Creative Team, 2013: *Adobe Flash Professional CS6-Classroom in a Book,* Pearson.
3. AGI Creative Team, 2012: *Adobe Flash Professional CS6 Digital Classroom,* Wiley India.
4. Alex Micheal, 2013: *Animating with Flash8:Creative Animation Techniques,* Focal Press.

B.Voc. Multimedia (Graphics & Animation) Semester- 3rd

Paper Title: Dreamweaver

Paper Code: MGA-304 **Credits: 6**

Job Role: Web Programmer.

Objective: To expose the students to the concept of HTML lanuage editor. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Introduction: Explore the Dreamweaver Workspace, Features of Dreamweaver, Website Development Process, Adding a Folder to the Website, Creating the Homepage, Adding Pages to a Website, Developing a Webpage, Creating Head Content and Set Page Properties, Create, Import and Format Text, Adding Links to Webpages, Using the History Panel, Editing the Code. Adding Image, Enhancing the Image and Use Alternate Text, Creating a Website Color Palette.

UNIT-II

Cascading Style Sheets: Introduction, Types, CSS Styles, Panel, Advantages & Disadvantages of CSS, Adding Rules, Creating Pages Using CSS Layouts.

Links: External& Internal Links, Create, Modify and Copy Spry Menu Bar, Create an Image Map, Tables, Creating a Table, Properties, Merging Cells, Insert &Align Images in a Table, Insert Text & Format Cell Content.

UNIT-III

Forms: Plan and Create Forms, Edit & Format the Forms, Form Objects, Test & Process Forms.

Positioning Objects with AP Div:Insert an AP Div, Position and Size, Adding Content to an AP Div, Adding Images, AP Div Properties, Using AP Elements Panel, Cloak Files, Check Files Out and In.

UNIT-IV

Media and Interactivity:Adding &Modifying Flash Objects, Inserting Rollover Images, Behaviors Panel, Adding Video and Sounds, Creating Pages Using Templates, Editable Regions, Optional Regions.

References:

1. Sherry Bishop, 2012: *Adobe Dreamweaver CS6 Reveald*, Cengage Learning.
2. Adobe Creative Team, James J. Maivald, 2012: *Adobe Dreamweaver CS6 Classroom in a Book*,
3. Adobe Press.
4. Jeremy Osborn, AGI Creative Team, 2012: *Adobe Dreamweaver CS6 Digital Classroom*,
5. Vickie Ellen Wolper, 2011: *Artistic Web Design Using Adobe Dreamweaver and Photoshop,*
6. Jones & Brelett Learning.
7. Patricia Robertson, 2014: *Dreamweavers*, Dreamweaver Press.

B.Voc. Multimedia (Graphics & Animation) Semester- 3rd

Paper Title: Practical Based on MGA-303 & MGA-304

Paper Code: MGA-305 **Credits: 6**

Job Role: 2D Animator, Web designer / Programmer.

There will be a practical examination in this paper along with semester examination. External examiner will evaluate the skills of the students in the Subject of Adobe Flash and Dreamweaver. Examiner will give on-the-spot assignments to the students. Besides this a viva voce (oral examination) of each student one by one will also be taken by the examiner for testing the knowledge of the students.

B.Voc. Multimedia (Graphics & Animation) Semester- 4th

Paper Title: ENVIRONMENTAL STUDIES

Paper Code: GEN-401 **Credits: 6**

Job Role: Environment Gaurd

Objective: The objective of the paper is to understand the basic concepts of the natural environment, to identify the different sources of pollution and their related effects, and to have awareness about the conservation strategies and laws followed in India.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Basics of environment: Introduction,concept of biosphere—lithosphere, hydrosphere, atmosphere. Ecology, types of ecosystems, natural resources-types, importance, conservation.

UNIT II

Renewable and non-renewable resources. Concept of sustainable development. Biodiversity: definition, hotspots, national parks, biosphere reserves, wildlife sanctuaries.

UNIT III

Pollution: Air, water, soil, noise and marine, Natural and anthropogenic sources of pollution. Effects of different types of pollutions Primary and secondary pollutants. Radioactive and thermal pollution. Acid rain.

UNIT IV

Global environment issues: Ozone depletion, global warming and climatic change. Environmental conservation strategies and projects in India, Ganga action plan, Yamuna action plan, Project tiger etc. Environmental laws in India.

Books Recommended:

- 'Ecology - Principles and Applications' by Chapman and Reiss, Cambridge University Press, Cambridge, U.K., 2004
- 'Basic Environmental Biotechnology' by G.E. Omenn, Plenum Press, N.Y., 1987.
- 'Environmental Challenges and Solutions' by Robert Jr. Cabin.
- 'Global Environmental Change and Land Use' by H.J.T.M Harmaan, Jan Verhagen, C.A. Rovers, Springer, 2003.
- 'A Textbook of Environment Science' by Thakur Vidya, Scientific Publishers, 2012.
- 'Living in the Environment : Principles, Connections, and Solutions' by Jr., G. Tyler Miller, Brooks Cole.

B.Voc. Multimedia (Graphics & Animation) Semester- 4th

Paper Title: Animation Theory

Paper Code: GEN-402 **Credits: 6**

Job Role: Scriptwriter.

Objective: The objective of the paper is to understand the basic concepts of the stroy, Scriptwriter and cameraman.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Development: Idea Creation, Evolving a Storyline, Scriptwriting.Character Design: Evolution of 2D and Evolution of 3D Character design, Animation Style, Concept and Environment Design. Project Financing: Animation Markets, Scheduling and Budgeting, Investment, Marketing and Distribution Possibilities. Short and Independent Film Developing.

UNIT-II

Rules of Filmmaking: Camera Positions, Camera Lenses, Lighting and Filters, Camera Moves, Staging, Scene-to-Scene Transitions, Screen Aspect Ratios. Soundtrack Recording and Editing: Talent Selection.

Storyboarding and Animatics: Storyboards, The Animatic.

UNIT-III

Digital Desktop Production: Stages of Animation Production, Production Team and Workflow, Project Management. Principles of Animation: Key Poses, Breakdowns and Inbetweens, Timing, Extreme Positions, Arcs and Path Action, Holds, Emphasis, Anticipation, Weight and Weighted Movements, Overlapping Action, Generic Walks, Walk cycles, Runs and Run Cycles, Dialogue and Lip Sync, Laughter, Takes.

Animating Step by Step: Key Poses, Drawing for Animators.

UNIT-IV

2D Animation Overview: It's All About Pencil and Paper, Tools of the Trade. 2D Animation Basics: Keys, Inbetweens and Timing, Dope Sheets & Production Folders, Flipping & Peg Bars. Finessing 2D Animation: Tracebacks, Eccentric Movement and Staggers, Panning & Camera Moves, Pan Speed & Strobing roblems, Shadows and Effects. 2D Vector Animation: Basic Approach, Vector Film Production, Being Resourceful, Non-Web Vector Animation. The Paperless Animation Studio: The Technology, Importance of Drawing

References:

1. Tony White, 2006: *Animation: From Pencil to Pixels*, Focal Press.
2. Isaac V. Kerlow :The Art of 3D Computer Animation and Effects, John Wiley & Sons.
3. Mary Murphy, 2008 :Beginner's Guide to Animation: Everything you need to know to get started, Watson-Guptill Publications.
4. Yueting Zhuang, Yunhe Pan, Jun Xiao, 2008: *A Modern Approach to Intelligent animation: Theory and Practice*, Springer Publications

B.Voc. Multimedia (Graphics & Animation) Semester- 4th

Paper Title: 3ds Max: Modeling

Paper Code: MGA-403 **Credits: 6**

Job Role: 3D Modler.

Objective: To expose the students to the concept of 3D Modler. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Exploring the Interface: Interface Elements, Menus, Toolbars, Viewports, Command Panel, Lower Interface Bar Control, Workspaces. Controlling & Configuring the Viewports: 3D Space, Navigation Gizmos, Using the Viewport Navigation Controls, Changing Viewport Display.

Working with Files: 3ds Max Scene Files, Importing and Exporting, File Utilities, Accessing File Information.

UNIT II

Setting Preferences: General, File, Gamma & Look-Up Table, Rendering & Radiosity, Animation, Kinametic, Gizmo, Mental Ray, Global Container Preferences.

Primitive Objects: Selecting System Units, Creating Primitive Objects, Exploring the Primitive Object Types, Architecture Primitives, Modeling Helpers.

Selecting Objects & Using Layers: Selecting Objects, Object Properties, Hiding & Freezing Objects, Using Layers, Scene Explorer, Object Properties, Display Properties.

UNIT III

Transforming Objects: Translating, Rotating & Scaling Objects, Transformation Tools, Pivot Points, Align Commands, Grids, Snap Options.

Cloning & Creating Object Arrays: Cloning Objects, Cloning Options, Mirroring Objects, Cloning Over Time, Creating Object Arrays.

Organizing Scenes: Containers, Referencing External Objects, File Link Manager, Schematic View Window, Hierarchies.

UNIT IV

Subobjects & Modifiers: Model Types, Normals, Subobjects, Modifiers, Modifiers Stack.
2D Splines & Shapes: Drawing in 2D, Editing in Splines.
Modelling with Polygons: Poly Objects, Creating Editable Poly Objects.
Graphite Modelling Tools, Freeform Tools, Selection Tools, Object Paint Tools.
Compound Objects: Types, Morphing Objects, Terrain Objects, Blob Mesh Objects, ProBoolean & Procutter Objects.

References:

1. Kelly L., Murdock, 2013 : *Autodesk 3Ds Max 2014 Bible*, Wiley.
2. Dariush Derakhshani, Randi L. Derakhshani, 2011: *Introducing 3ds Max 9-3D for Beginners,* Wiley India.
3. Kelly L. Murdock, 2013:*Mastering Autodesk 3ds Max 2013*, Wiley India.
4. Markus Kuhlo, 2013: *Architectural Rendering with 3ds Max and V-Ray*, Focal Press.
5. Todd Daniele, 2009: *Poly-modeling with 3ds Max: Thinking Outside of the Box, Focal Press.*

B.Voc. Multimedia (Graphics & Animation) Semester- 4th

Paper Title: 3ds Max: Texturing, Lighting & Rendering

Paper Code: MGA-404 **Credits: 6**

Job Role: 3D Virtual Envirnmental Creator

Objective: To expose the students to the concept of 3D Virtual Envirnmental Creator. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Applying Textures: Understanding Material Properties, Working with State Material Editor, Using Shading Types, Using Specialized Material Types, Creating Compound Materials, Using Matt/Shadow Material.

Unwrapping UVS and Mapping Structure: Mapping Modifiers, Using the Unwrap UVW Modifier, Creating Baked Structure and Normal Maps.

UNIT-II

Working with Backgrounds Cameras and Lightening: Configuring and Aiming Cameras, Setting Camera Parameters, Camera Connection Modifier, Loading a Background Image and Camera.

Using Lights and Basic Lighting Techniques: Understanding the Basic of Lights, Types of Lights, Creating and Positioning Light Objects, Viewing a Scene from Light.

UNIT-III

Using Sunlight and Daylight System: Introduction of Sunlight and Daylight System, Understanding Azimuth and Altitude, Working with Light Tracing, Understanding Radiosity, Using Advanced Lighting Material.

Rendering:Working with Render Parameters, Adding Pre-Render and Post-Render Scripts, Rendering Stylized Scenes, Managing Render States.

UNIT-IV

Batch and Network Rendering: Batch Rendering Scenes, Understanding Network Rendering, Setting up a Network Render System, Setting up 3ds Max .

Rendering with Mental Ray and Iray: Enabling Mental Ray and Iray, working with Mental Ray and Iray, using Render Elements.

References:

1. Kelly L., Murdock, 2013 : *Autodesk 3Ds Max 2014 Bible*, Wiley.
2. Dariush Derakhshani, Randi L. Derakhshani, 2011: *Introducing 3ds Max 9-3D for Beginners,* Wiley India.
3. Kelly L. Murdock, 2013:*Mastering Autodesk 3ds Max 2013*, Wiley India.
4. Markus Kuhlo, 2013: *Architectural Rendering with 3ds Max and V-Ray*, Focal Press.
5. Todd Daniele, 2009: *Poly-modeling with 3ds Max: Thinking Outside of the Box, Focal Press.*

B.Voc. Multimedia (Graphics & Animation) Semester- 4th

Paper Title: Practical Based on Paper MGA-403 & MGA-404

Paper Code: MGA-405 **Credits: 6**

Job Role: 3D Virtual Environmental Creator.

There will be a practical examination in this paper along with semester examination. External examiner will evaluate the skills of the students in the field of animation using 3ds Max. Examiner will give on-the-spot assignments to the students. Besides this a viva voce (oral examination) of each student one by one will also be taken by the examiner for testing the knowledge of the students.

B.Voc. Multimedia (Graphics & Animation) Semester- 5th

Paper Title: CRITICAL THINKING AND ELEMENTARY SATISTICS

Paper Code: GEN-501 **Credits: 6**

Job Role: Satistical Analyst

Objective: The objective of the paper is to provide knowledge to student about basics of statistics and to inculcate the habit of critical thinking.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit I

Introduction: Statistics as a Subject ; Statistical Data: Meaning and Types , Collection and Rounding of Data, Classification and Presentation of Data.

Unit II

Analysis of Univariate Data: Construction of a Frequency Distribution; Concept of Central Tendency, mean, median and mode.

Unit III

Dispersion and Their Measures; Time Series: Meaning, Components, Models, Fitting Linear and Quadratic Trend

Unit IV

Index Number: Meaning, Types, and Uses: Methods of Constructing Price and Quantity Indices (Simple and Aggregate).

Books Recommended:

1. Aczel, Amir D.: Complete Business Statistics, Irwin.
2. Chou-YaLun: Statistical Analysis, Holt, Rinchart and Winslon.
3. Levin, Richard I. & David S Rubin: Statistics for Management, Prentice Hall of India, Delhi.

B.Voc. Multimedia (Graphics & Animation) Semester- 5th

Paper Title: Cyber Laws

Paper Code: MGA-502 **Credits: 6**

Job Role: Cyber Crime Controller.

Objective: The subject and the course content will help the student to understand, explore and acquire a critical understanding of Cyber Laws.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Introduction to Cyber Laws – In Indian Context: - Information Technology Act 2000, Amendments made in the Indian ITA 2000, Positive Aspects of the ITA 2000, The Weak Areas of the ITA 2000, Challenges to Indian Law and Cybercrime, Scenario in India, Consequences of Not Addressing the Weakness in Information Technology Act Amendments to the Indian ITA 2008, Impact of IT Act Amendments Impact Information Technology Organizations, Cybercrime and Punishment.

UNIT II

Internet and the Protection of Software Copyright: - Open Source, Reverse Engineering Trademark Issues in Cyber Space: - Domain Name, the ICANN Uniform Domain, Name Dispute Resolution Policy

UNIT III

IPR in Cyber Space: - Patents in Digital Technology, Copy Rights in Digital Space, WIPO Internet Treaties, Trademark Online, IP Related Cyber Crimes: - Introduction, Essential Ingredients of Crime, Types of Internet Crimes, Cyber Crime and IPR

UNIT IV

Indian Contract Act 1872:- Essential elements of a valid Contract, Types of Contract, Performance of Contract, Discharge of Contract. Licensing Agreement: - Benefit of Licensing, Licensing of the Basic Intellectual Property.

References:

1 Chander, 2012: *Cyber Laws and IT Protection*, Harish Publisher: PHI, New Delhi.
2 Adam J. Mambi, 2010: *ICT Law Book:A Source Book for Information and Communication Technologies,* Mkuki na Nyota Publishers.
3 Jonathan Rosenoer, 1997: *Cyber Law: The Law of Internet*, Springer Publishers.

B.Voc. Multimedia (Graphics & Animation) Semester- 5th

Paper Title: 3D Max Animation (Practical)

Paper Code: MGA-503 **Credits: 6**

Job Role: 3D Animator.

Objective: To expose the students to the concept of 3D Animation. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Unit-1

Forest Scene in 3DS Max:

- Preparing the Scene
- Camera Setup
- Creating Light Sources
- Texture the Scene
- Fine Tuning
- Final Render Settings

Unit-2

Environment Modeling:

- Preparing the Scene
- Camera Setup
- Lighting & Texturing
- Final Render Settings

Unit-3

Bedroom, Kitchen, Bathroom Interior Design:

- Preparing the Scene
- Camera Setup
- Create Light Sources
- Texture the Scene
- Fine-Tuning
- Final Render Settings

Unit-4

House Exterior:

- *Preparing the Scene*
- Create Additional Materials
- Light Inside
- HDRI Illumination
- Fine-Tuning Shadows on the facade
- Final Render Settings

Text Books

1. Name of Authors Title of the Book Publisher
2. 3dsmax7 Fundamentals New Riders
3. TedBoardman 3d'sMax5Fundamentals Tech media
4. Inside 3ds max7 NewRiders
5. Michelebousquet Modelrig,Animate with 3d's max6 Many world production

Reference Books

1. Michael E. Mortenson 3D Modeling, Animation, and Rendering Create space
2. Boris Kulagin 3ds Max 8 from Modeling to Animation Bpb
3. Michael G. 3D Modeling and Animation Igi Publishing

B.Voc. Multimedia (Graphics & Animation) Semester- 5th

Paper Title: Visual Effects (Practical)

Paper Code: MGA-504 **Credits: 6**

Job Role: Visual Effect Artist

Objective: To expose the students to the concept of Visual Effect Artist. To fulfill the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world.

Note:-
- Composition, visual aesthetics, technical proficiency, and artistic merit **50%**
- Completeness of project and clear communication of a unified concept **50%**

Unit-I

Self Portrait (Chroma Key) - (Photoshop, after effects) Using media generated in class and images and video created and collected from the first homework assignment, compose a brief (30 seconds) self portrait video. Use any stills, moving images, titles, effects, music or sound that you wish. Final product must include: sound, animated stills, applied effects, chroma key video, transitions, and titles.

Unit-II

Title Sequence (as 3D motion graphic choreography) – (after effects, Photoshop, illustrator)
Create a title sequence for an invented television show or movie. Final version should be 30 seconds to 2 minutes long, utilize animated and/or processed text with layers of moving images. The final piece may be abstract or representational in its presentation. The project should emphasize 3D space, composition, design, and type in motion**.**

Unit-III

Film Festival Show Opener – (after effects, final cut, photoshop, etc…)
Create a concept for an opening sequence to a film festival (either the EMF Media Arts Festival or another festival of your choice). The opener should be 30 seconds-1 minute in length and is open ended in terms of content, but it must utilize digital tools and involve 3D compositing, lighting, and camera movement in After Effects.

Unit-IV

DVD and Demo Reel – (DVD studio Pro and any other software as needed) Compile all of your video work from the class onto DVD. The interface of the DVD must be original (NO templates), with menus,
Transitions, and navigation all from scratch. Along with each project compiled on the DVD you must also edit together a short 30-second reel showcasing the highlights of your motion graphic work from the semester.
The 30-second highlight reel should be included on the compilation DVD and also compressed and uploaded to either google video, youtube, vimeo, or another similar site.

Design a DVD cover and disc label. The graduate students are expected to also include work from outside of the class in the Demo Reel and must utilize animated transitions/effects for DVD menus and button transitions.

Books Recommend:

- Adobe After Effects CC Visual Effects and Compositing Studio Techniques by Mark Christiansen (Author).
- Compositing Visual Effects in After Effects: Essential Techniques by Lee Lanier (Author).

B.Voc. Multimedia (Graphics & Animation) Semester- 5th

Paper Title: PRE PRODUCTION (THEORY)

Paper Code: MGA-505 **Credits: 6**

Job Role: Pre-production refers to the tasks that must be completed or executed before filming or shooting begins. This includes tasks such as hiring actors or models, building sets, budgeting, planning, scheduling, renting equipment and tests, to name a few of the many pre-production tasks.

Objective: The goal of pre-production is to develop an efficient structure for your project on which the final animation will be produced. In pre-production, we identify potential difficulties in production and work with you to minimize or eliminate them.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT: 1

Production Philosophy: The Production Process, Importance of Goals-Audience Analysis, Importance of Workflow Difference between Studio and Field Production, Importance of Technology.

UNIT: 2

The Technology: Connecting the Real World to the Digital World, Audio Signals, Video Signals Synchronization, Audio and Video Compression
The Equipment: Camera**,** Viewfinder and Camera Controls , Camera Supports, Lighting.

UNIT: 3

Preproduction: Introduction, Script Formats Organizing Forms.
Production: Production Stages and Setup, Setting Up Lighting Preparation Shooting, Framing Principles, Creating Movement and Editing.

UNIT: 4

Postproduction: Soul of Production, Editing Concept, Process Background, Hardware, Software, Physical and Technical Process, The Aesthetic Process, Output Distribution- Preparing to Output/Export ,Output Process.

Suggested books for Reading:

- The Five C's of Cinematography: Motion Picture Filming Techniques , by Joseph V. Mascelli
- Single Camera Video Production, by Robert B Musburger.

B.Voc. Multimedia (Graphics & Animation) Semester- 6th

Paper Title: Entrepreneurship Development Programme

Paper Code: GEN-601 **Credits: 6**

Job Role: Entrepreneur

Objective: EDPs aim at training various target groups in entrepreneurial traits so that they obtain adequate information, motivation and guidance in setting up their own enterprises.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-1

Entrepreneur: Definition, nature, and characteristics of entrepreneurship: Emergence of entrepreneurial class including women entrepreneurs: Theories of entrepreneurship: Socioeconomic environment and the entrepreneur. Characteristics of Entrepreneur Leadership, Risk taking, Decision making and business planning. Innovation and Entrepreneurship, entrepreneurial behaviour and motivation.

UNIT-II

Small Business as a Seed Bed of Entrepreneurship: Concept of business venture. The start-up process, Concept, Plan, Implementation, Initial Strategic Planning, Product and Marketing Scope, Legal and Tax consideration, Risk analysis and financial considerations. Profit Planning in Small Enterprise: Growth Strategies and diversification.

UNIT-III

Finance Management in current operations and expansion of capital. Role of Small Business in the national economy, National Policies for small business development, Governmental and Non-Governmental assistance. Contribution of Commercial Banks in promoting and servicing small business. Small business and modern technology.

UNIT-IV

Rules & Regulations: Governmental Policies and formalities in setting up a unit. Basic requirements regarding registration, Excise, Sales Tax, Factory Act, SSI exemptions: Licensing and Registration procedure; Appreciation of important provisions of Factory Act, Shops & Commercial Establishment Act; Sales of Goods Act, Partnership Act; Contract Act; Income Tax, Sales Tax and Excise rules; Insurance.

References

1. Desai, Vasant (2003). Small-Scale Industries and Entrepreneurship. Himalaya Publishing House, Delhi.
2. Kaulgud, Aruna (2003). Entrepreneurship Management. Vikas Publishing House, Delhi.
3. Cynthia, L. Greene (2004). Entrepreneurship Ideas in Action. Thomson Asia Pvt. Ltd., Singapore.
4. Chandra, Ravi (2003). Entrepreneurial Success: A Psychological Study. Sterling Publication Pvt. Ltd., New Delhi.
5. . Balaraju, Theduri (2004). Entrepreneurship Development: An Analytical Study. Akansha Publishing House, Uttam Nagar, New Delhi.
6. David, Otes (2004). A Guide to Entrepreneurship. Jaico Books Publishing House, Delhi.
7. Taneja (2004). Entrepreneurship. Galgotia Publishers

B.Voc. Multimedia (Graphics & Animation) Semester- 6th

Paper Title: Design & Communication Process

Paper Code: GEN-602 **Credits: 6**

Job Role: Graphics Designing.

Objective: To develop intellectual awareness and curiosity to acquire the knowledge necessary to sustain effective work in visual communication design.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit-I

Basic Principles of composition: What is composition, why learn these principles, fulfilling the purpose.
The Subject: Selection and study, drawing steps.

Unit- II

Three Principles: Unity, Balance, Center of interest.
Achieving Emphasis: Light and shad, Details, contrasts.

Unit-III

Six Visual elements: - Color, Line, Shape, Value & Tone, Texture, Volume.
Pictorial principles:- Contrast, Harmony, Emphasis, Movement, Proportion, Pattern, Variety, Rhythm, Repetition.

Unit-IV

Balance: Asymmetrical balance, informal balance, radial balance.

Text books: -

- Design and Communication: A Foundation Course By Peter Gowers ,Blackie, 1988 pub.(UK)

Reference book: -

- Design for communication: conceptual graphic design basics By Elizabeth Resnick ,John Wiley & Sons pub.(India)

B.Voc. Multimedia (Graphics & Animation) Semester- 6th

Paper Title: FINAL PROJECT

Paper Code: MGA-603 **Credits: 6**

Job Role: Graphics Designer, Web Designer, Video / Audio Editing, VFX Compositor, 2D Animator, 3D Modular.

Specialization (Choose any one Project)

Final Project: Graphics Designing / Web Designing / 2D Animation / 3D Animation / Texturing, Shading & Lighting, Rendering / Visual Effects / Digital Compositing / Motion Graphics / Video Editing.